



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE (Advanced)

FULL SYLLABUS

16-05-2023

JEE(Main + Advanced) : ENTHUSIAST COURSE /ALL STAR BATCH/(SCORE ADVANCED)

ANSWER KEY

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,C	A,C,D	B,D	A,B,D	A,C,D	C,D		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	C	C	B	A				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	4.00	1.46	0.19	4.00	2.10 to 2.14	3.13 to 3.17	1.60	0.75

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,B,C,D	C,D	B,C	A,B,C	A,B,D	B,C,D		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	D	A	A	A				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	0.18 to 0.22	8	3	3.5	18	19	0	1

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,B,C,D	A,B	A,C,D	A,C,D	A,B,C,D	A,C		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	A	B	C	D				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	87	17.12 or 17.13	3	2.23 or 2.24	2.14	7	16	4

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (A,C)

Let the impulse due to the rod be T.

$mv - Tsina = 3mv_1$ where, V_1 is the speed of (A + C) just after the collision.

$Tcosa = 3mV_2$ where V_2 is the speed of B from the inextensibility of rod AB

$$V_1 \sin \alpha = V_2 \cos \alpha$$

$$\text{Thus, } V_1 = \frac{v \cos^2 \alpha}{3} \text{ \& } V_2 = \frac{v \sin^2 \alpha}{6}$$

2. Ans (A,C,D)

$$\text{Centrifugal force} = \frac{dU}{dr} = kr = \frac{mv^2}{r},$$

$$\text{Angular momentum, } L = mvr = \frac{nh}{2\pi}$$

On solving, we get

$$r = \sqrt{\frac{nh}{2\pi\sqrt{km}}} \text{ \& } v = \sqrt{\frac{nh}{2\pi m}} \sqrt{\frac{k}{m}};$$

$$KE = \frac{1}{2}mv^2 = \frac{nh}{4\pi} \sqrt{\frac{k}{m}}$$

$$U = \frac{kr^2}{2} = \frac{k}{2} \left(\frac{nh}{2\pi\sqrt{km}} \right)$$

$$= \frac{nh}{4\pi} \sqrt{\frac{k}{m}};$$

$$E = KE + U$$

$$= \frac{nh}{2\pi} \sqrt{\frac{k}{m}}$$

3. Ans (B,D)

$$\text{Flux through frame} = \frac{\mu_0 i}{2\pi} \int_2^4 \frac{4dx}{x}$$

$$\therefore \phi = \left(\frac{2\mu_0 \ell \ln 2}{\pi} \right) i$$

$$\therefore \varepsilon = \left| \frac{d\phi}{dt} \right| = \left(\frac{2\mu_0 \ell \ln 2}{\pi} \right) \frac{di}{dt}$$

$$= \frac{2\mu_0 i_0 \omega \ell \ln 2}{\pi} \cos \left(\omega t + \frac{\pi}{6} \right)$$

$$\Rightarrow Q = \xi C = \left(\frac{2\mu_0 i_0 \omega \ell \ln 2}{\pi} \right) \cos \left(\omega t + \frac{\pi}{6} \right) C$$

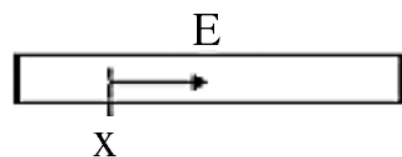
$$\therefore Q_{\max} = \varepsilon_{\max} C = \left(\frac{2\mu_0 i_0 \omega \ell \ln 2}{\pi} \right) C$$

and

$$i_{\text{frame}} = \left| \frac{dq}{dt} \right| = \left(\frac{2\mu_0 i_0 \omega^2 \ell \ln 2}{\pi} \right) \left(\omega t + \frac{\pi}{6} \right)$$

4. Ans (A,B,D)

Let at any point E be field & intensity. Then
Ohm's Law



$$E = P \cdot J = \frac{P_0}{1+x} \cdot \frac{1}{\text{area}}$$

$$E = \frac{P_0}{1+x} \cdot I \quad \dots\dots\dots(i)$$

$$E \cdot dx = -dV \Rightarrow -dV = \frac{P_0 I}{1+x} \cdot dx$$

$$-\int_A^B dV = P \cdot I \int_0^1 \frac{dx}{1+x}$$

$$= P_0 I [\ln(1+x)]_0^1 = P_0 I \cdot \ln 2$$

$$V_A - V_B = (P_0 \ln 2) I \Rightarrow V = I \cdot (P_0 \ln 2)$$

$$\text{So } R_0 = \frac{V}{I} = P_0 \ln 2,$$

$$E = \frac{1}{1+x} \cdot \frac{V}{\ln 2}, E_A = \frac{V}{\ln 2} E_B = \frac{V}{2 \ln 2},$$

$$E_C = \frac{2V}{3 \ln 2}$$

5. Ans (A,C,D)

$$y = \left(\frac{\lambda \cdot D}{d} \right) \cdot n \Rightarrow n = \frac{y \cdot d}{\lambda} (D)$$

$$n = 1, 2, 3, 4, \dots\dots \text{ for maxima}$$

$$n = 0.5, 1.5, 2.5, \dots\dots \text{ for minima}$$

$$D = x + vt \quad n = \frac{dy}{\lambda} / [x + vt]$$

$$\text{At } t = 0, n = 4 \Rightarrow 4 = \left(\frac{yd}{\lambda} \right) / x$$

$$4x = \frac{yd}{\lambda} \Rightarrow n = \frac{4x}{x + vt},$$

$$t = \frac{x}{v}, n = \frac{4x}{2x} = 2$$

$$t = \frac{5x}{3v}, n = \frac{4x}{x + \frac{5x}{3}} = \frac{12x}{8x}$$

$$n = 1.5$$

6. Ans (C,D)

$$\frac{dA}{dt} = \frac{L}{2m}$$

$$KE_i = \frac{GMm}{2r}$$

$$U_i = \frac{GMm}{r}$$

$$TE_i = -\frac{GMm}{2r}$$

$$KE_f = \frac{6}{5} \frac{GMm}{2r}$$

$$V_f = \sqrt{\frac{6}{5} \frac{Gm}{r}}$$

MEC

$$\frac{1}{2} m \frac{6}{5} \frac{GM}{R} - \frac{Gmm}{R}$$

$$= \frac{1}{2} m v_a^2 - \frac{GMm}{r_a}$$

AMC

$$m \sqrt{\frac{6}{5} \frac{Gm}{R}} R = m v_a r_a$$

PART-1 : PHYSICS

SECTION-I (ii)

7. Ans (C)

By angular momentum conservation about the hinge,

$$L_i = L_f$$

$$\therefore mv \frac{L}{2} = \frac{mL^2}{3} \omega$$

$$\therefore \omega = \frac{3v}{2L} : t = \frac{2L}{3v}$$

$$v_{CM} = \frac{\omega L}{2} = \frac{3v}{4}$$

$$KE_f = \frac{1}{2} I \omega^2 = 25 \frac{mv^2}{L}$$

$$= \frac{1}{2} \times \frac{mL^2}{3} \times \left(\frac{3v}{2L} \right)^2$$

$$\text{Later, } F_{\text{hinge}} = m\omega^2 \frac{L}{2} = \frac{3}{8} mv^2$$

$$\text{Impulse} = m \left(v - \frac{3v}{4} \right)$$

$$= \frac{mv}{4} = m \left(\frac{3v}{2L} \right)^2 \times \frac{L}{2}$$

$$\therefore F_{\text{hinge}} = \frac{mv}{4} \div \frac{L}{100v} = \frac{9mv^2}{8L}$$

8. Ans (C)

$$F_1 = \left(\frac{a}{4} \times \rho \times g \right) \times \left(a \times \frac{a}{2} \right)$$

$$= \frac{1}{8} a^3 \rho g$$

$$F_2 = \left(\frac{3}{4} a \rho g \right) \times \left(a \times \frac{a}{2} \right)$$

$$= \frac{3}{8} a^3 \rho g$$

$$F_3 = \left(\frac{2}{3} a \rho g \right) \times \left(a \times \frac{a}{2} \right)$$

$$= \frac{1}{3} a^3 \rho g$$

$$F_4 = \left(\frac{1}{3} a \rho g \right) \times \left(a \times \frac{a}{2} \right)$$

$$= \frac{1}{6} a^3 \rho g$$

9. Ans (B)

$$(i) E = \frac{KQ}{d^2} \Rightarrow E \propto \frac{1}{d^2}$$

(ii) Dipole

$$E = \frac{2kp}{d^3} \sqrt{1 + 3\cos^2\theta}$$

$$E \propto \frac{1}{d^3} \text{ for dipole}$$

(iii) For line charge

$$E = \frac{2k\lambda}{d}$$

$$E \propto \frac{1}{d}$$

$$(iv) E = \frac{2K\lambda}{d - \ell} - \frac{2K\lambda}{d + \ell}$$

$$= 2K\lambda \left[\frac{d + \ell - d + \ell}{d^2 - \ell^2} \right]$$

$$E = \frac{2K\lambda(2\ell)}{d^2 \left[1 - \frac{\ell^2}{d^2} \right]}$$

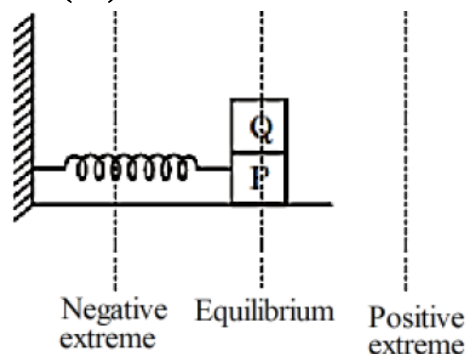
$$E \propto \frac{1}{d^2}$$

(v) Electric field due to sheet

$$E = \frac{\sigma}{2\epsilon_0}$$

$E = v$ is independent of r

10. Ans (A)



Friction on Q will always be towards equilibrium and friction on P will be opposite to that friction is only force acting on block Q in horizontal direction which is responsible for its oscillation.

Friction force will be maximum at extreme position and zero at equilibrium.

PART-1 : PHYSICS

SECTION-II

1. Ans (4.00)

$$\text{Area of } \triangle ABC = \frac{1}{2} (2.5 \times 10^5) (0.05 \times 10^{-3})$$

$$= \frac{125}{2} = \frac{25}{4}$$

Since, $\triangle AEB$ and $\triangle BCD$ are similar, therefore

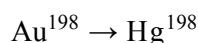
$$\frac{1.5}{2.5} = \frac{CD}{0.05} \Rightarrow CD = \frac{3}{5} \times 0.05 = 0.03$$

$$\text{Area of } \triangle BCD = \frac{1}{2} \times (1.5 \times 10^5) (0.03 \times 10^{-3})$$

$$= \frac{4.5}{2} = \frac{9}{4} \text{ W}_N = \frac{25}{4} - \frac{9}{4} = 4 \text{ J}$$

2. Ans (1.46)

Total energy released from



$$\text{In ground state} = (\Delta m_{\text{loss}}) c^2$$

Energy released from ^{198}Hg in first

Existed state $\rightarrow \text{Hg}$ in ground state $= (-1.6) - (-2) \text{ MeV}$

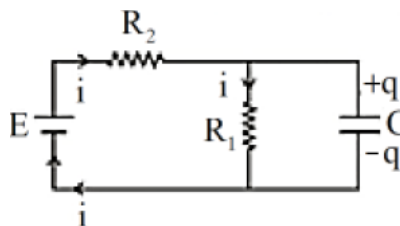


Second excited state

$$= 1.86 - 0.4 = 1.46 = \text{max. K.E. of } \beta \text{ particle}$$

3. Ans (0.19)

Capacitor has maximum charge at steady state



$$\frac{q}{C} = iR_1$$

$$q = \frac{CER_1}{R_1 + R_2}$$

$$\ln q = \ln C + \ln E + \ln R_1 - \ln(R_1 + R_2)$$

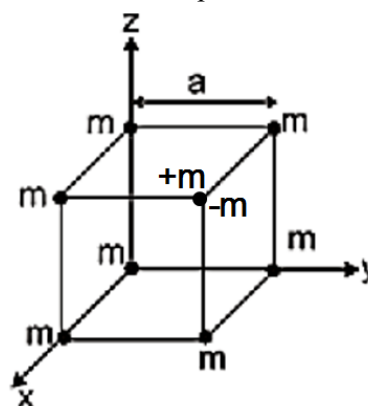
$$\frac{\Delta q}{q} = \frac{\Delta C}{C} + \frac{\Delta E}{E} + \frac{\Delta R_1}{R_1} - \left(\frac{\Delta R_1 + \Delta R_2}{R_1 + R_2} \right)$$

$$\frac{\Delta q}{q} = \frac{\Delta C}{C} + \frac{\Delta E}{E} + \Delta R_1 \left\{ \frac{1}{R_1} - \frac{1}{R_1 + R_2} \right\} - \frac{\Delta R_2}{R_1 + R_2}$$

$$\left| \frac{\Delta q}{q} \right|_{\text{max}} = \frac{0.01}{1} + \frac{4}{40} + 3 \left\{ \frac{1}{60} - \frac{1}{90} \right\} + \frac{6}{90}$$

4. Ans (4.00)

Place +m and -m particle at the vacant corner



Assume the +m particle and the rest seven particle as system(1)

The total mass of system (1) is $m_1 = 8m$ and the position vector of its com is

$$\vec{r}_1 = \frac{a}{2} \hat{i} + \frac{a}{2} \hat{j} + \frac{a}{2} \hat{k} \text{ and assume } -m \text{ particles system (ii)}$$

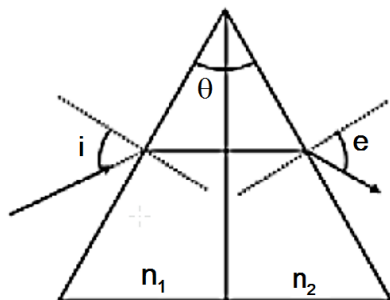
$$m_2 = -m, \vec{r}_2 = a\hat{i} + a\hat{j} + a\hat{k}$$

$$\vec{r}_{\text{cm}} = \frac{m_1 \vec{r}_1 + m_2 \vec{r}_2}{m_1 + m_2}$$

$$= \frac{(8m) \left(\frac{a}{2} \hat{i} + \frac{a}{2} \hat{j} + \frac{a}{2} \hat{k} \right) + (-m) (a\hat{i} + a\hat{j} + a\hat{k})}{8m + (-m)}$$

$$\vec{r}_{\text{cm}} = \frac{3a}{7} \hat{i} + \frac{3a}{7} \hat{j} + \frac{3a}{7} \hat{k}$$

5. Ans (2.10 to 2.14)



$$1 \times \sin i = \mu \sin\left(\frac{A}{2}\right),$$

$$\sin i = \frac{3}{4}$$

$$n_1 \sin 30^\circ = 1 \sin(e)$$

On differentiating both sides

$$dn \sin 30^\circ = de \cos(e)$$

$$de = \frac{dn}{2 \cos(e)}$$

$$= \frac{dn}{2 \times \frac{\sqrt{7}}{4}}$$

$$de = \frac{2}{\sqrt{7}} dn \Rightarrow de < dn$$

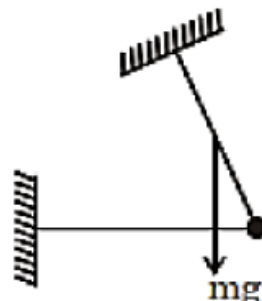
$$de = \frac{2.8 \times 10^{-3} \times 2}{\sqrt{7}} = 2.1166 \text{ mrad}$$

6. Ans (3.13 to 3.17)

$$T_1 = 2\pi \sqrt{\frac{m\ell^2/3}{mg\ell/2}} = 2\pi \sqrt{\frac{2\ell}{3g}}$$

$$T_1 = 2\pi \sqrt{\frac{m\ell^2}{3 \left(mg\frac{\ell}{2} + k\ell^2 \right)}}$$

$$= 2\pi \sqrt{\frac{m\ell}{3 \left(\frac{3mg}{2} \right)}}$$



$$- \left(mgx \frac{\ell}{2} \theta + k\ell\theta\ell \right) = \frac{m\ell^2}{3} \alpha$$

$$= 2\pi \sqrt{\frac{2}{9} \frac{\ell}{g}}$$

$$\alpha = \frac{\left(mgx \frac{\ell}{2} + k\ell^2 \right)}{\frac{m\ell^2}{3}} \theta$$

$$T = \pi \sqrt{\frac{2\ell}{3g}} + \pi \sqrt{\frac{2\ell}{9g}}$$

$$= \pi \sqrt{\frac{2\ell}{3g}} \left(1 + \frac{1}{\sqrt{3}} \right)$$

$$= 2(1+0.577) = 3.15 \text{ sec}$$

7. Ans (1.60)

$$f_1 = f_0$$

$$f_2 = \frac{c+v}{c-v} \times f_0$$

$$f_b = 256 \left(\frac{2v}{c-v} \right) = 1.6 \text{ Hz}$$

8. Ans (0.75)

$$\vec{F} = (\alpha y \hat{i} + 2\alpha x \hat{j})$$

$$W_{AB} = (-1\hat{i}) \cdot (1\hat{i}) = -1 \text{ J}$$

$$\begin{bmatrix} \vec{F} = -1\hat{i} + 2\alpha x \hat{j} \\ \vec{S} = 1\hat{i} \end{bmatrix}$$

Similarly,

$$W_{BC} = 1 \text{ J}$$

$$W_{CD} = 0.25 \text{ J}$$

$$W_{DE} = 0.5 \text{ J}$$

$$W_{EF} = W_{FA} = 0 \text{ J}$$

$$\therefore \text{New work in cycle} = 0.75 \text{ J}$$

PART-2 : CHEMISTRY

SECTION-I (i)

3. Ans (B,C)

B) Reactivity towards water $\uparrow$ from Li to Cs due to the $\downarrow$ melting points of metals

C) fact

D) KCl solubility in $H_2O = 27.77 \text{ g/100 mL } (0^\circ\text{C})$

KCl solubility in $H_2O = 36 \text{ g/100 ml pure water at}$

$T = 25^\circ\text{C}$

5. Ans (A,B,D)

10 mole A and 10 mole B

$$P_T = P_A^0 X_A + P_B^0 X_B$$

$$P_T = 0.6 \times \frac{1}{2} + 9 \times \frac{1}{2}$$

$$P_T = 0.75$$

Since $1/4^{\text{th}}$ mole will be converted

$\therefore$ In vapour 4 moles total will be present.

Assuming x mole to be of A.

	Liquid	Vapour
A	$10 - x$	x

B	$5 + x$	$5 - x$
---	---------	---------

$$(i) \frac{x}{5} = \frac{10 - x}{15} \times \frac{0.6}{P_T}$$

$$P_T = \frac{(10 - x) \times 0.6}{3x}$$

$$(ii) \frac{5 - x}{5} = \frac{5 + x}{15} \times \frac{0.9}{P_T}$$

$$P_T = \frac{(5 - x)}{3(5 - x)} \times 0.9$$

$$\Rightarrow \frac{(10 - x) \times 6}{3x} = \frac{(5 + x) \times 9}{3(5 - x)}$$

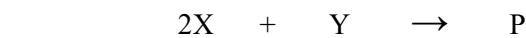
Simplifying $(x^2 + 45x - 100) = 0$

$$\Rightarrow x = \frac{-45 + \sqrt{2425}}{2} = 2.125$$

PART-2 : CHEMISTRY

SECTION-I (ii)

8. Ans (A)



$$t = 0 \text{ conc. } \quad 2 \quad 1 \quad -$$

$$t \text{ conc. } \quad 2 - 2x \quad 1 - x \quad x$$

$$t = 50 \text{ sec } n_Y = 0.5 \Rightarrow t_{1/2} = 50 \text{ sec}$$

$$A) 2K_{r \times n} t = \ln \left(\frac{X_0}{X_t} \right)$$

$$t_{1/2} = \frac{\ln 2}{2K_{r \times n}} = \frac{0.693}{2 \times 2K_{r \times n}} = 50$$

$$K_{r \times n} = 0.693 \times 10^{-2} = 6.93 \times 10^{-3} \text{ sec}^{-1}$$

(A) is correct

(B) is correct

$$(C) -\frac{d[X]}{dt} = 2K_{r \times n} [X]$$

$$= 2 \times 6.93 \times 10^{-3} \times 1$$

$$= 13.86 \times 10^{-3} \text{ M sec}^{-1} \text{ (correct)}$$

$$(D) -\frac{d[Y]}{dt} = K_{r \times n} [X]$$

9. Ans (A)

(P)

$$a \xrightarrow[=18 \text{ sec}]{t_{1/3}} \frac{2a}{3} \xrightarrow[=18 \text{ sec}]{t_{1/3}} \frac{4a}{9} \xrightarrow[=18 \text{ sec}]{t_{1/3}} \frac{8a}{27}$$

$t_{19/17} = 54 \text{ sec}$

(Q)

$$a \xrightarrow[=16 \text{ sec}]{t_{1/4}} \frac{3a}{4} \xrightarrow[=16 \text{ sec}]{t_{1/4}} \frac{9a}{16}$$

$t_{7/16} = 32 \text{ sec}$

$$(R) K = \frac{a - \frac{2a}{3}}{18} = \frac{x}{30}$$

$$\Rightarrow x = \frac{5}{9} a$$

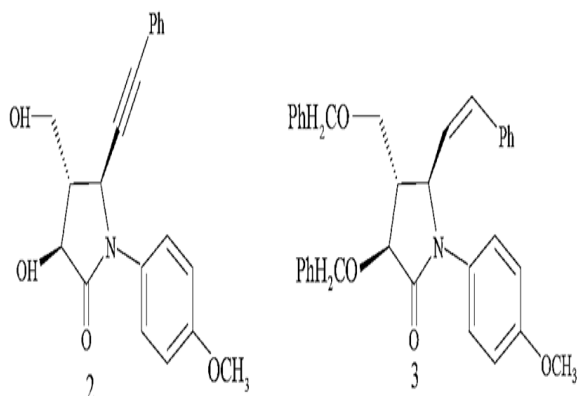
$$(S) K = \frac{a - \frac{a}{2}}{16} = \frac{x}{28}$$

$$\Rightarrow x = \frac{7}{8} a$$

PART-2 : CHEMISTRY

SECTION-II

1. Ans (0.18 to 0.22)



The molecular weight of final product is 519.

0.20 moles of reactant gives 50.6 gm of final product with the given yield.

2. Ans (8)

- (i) IF_5 (sp^3d) (ii) ICl_4^- (sp^3d^2)
 (iii) XeO_2F_2 (sp^3d) (iv) NH_2^- (sp^3)
 (v) BCl_3 (sp^2) (vi) BeCl_2 (sp)
 (vii) AsCl_4^+ (sp^3) (viii) $\text{B}(\text{OH})_3$ (sp^2)
 (ix) NO_2^- (sp^2) (x) ClO_2^+ (sp^2)

$$x = (\text{sp}^3)^2 + \text{sp}^3\text{d}(1) + \text{sp}^3\text{d}^2(2) = 5$$

$$y = 4, z = 1; 5 + 4 - 1 \Rightarrow 8$$

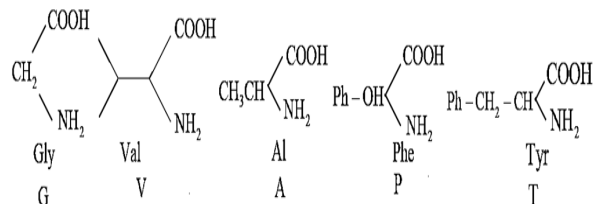
3. Ans (3)

$$x=0, y=3$$

4. Ans (3.5)

All give oxygen on heating.

5. Ans (18)



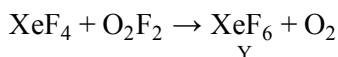
V G A P T V G P A T
 V A G P T V P G A T
 V A P G T V P A G T

6 combinations by

each of V, A, P

$\therefore$ Total 18 possible sequences.

6. Ans (19)



Y has 3 lone pair of electron in each fluorine and one lone pair of electron in xenon.

Hence total lone pair of electrons is 19.

7. Ans (0)

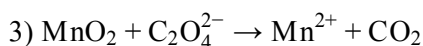
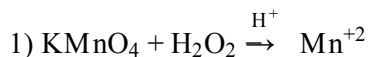
$$\Delta S_{\text{cycle}} = 0 = \frac{q_1}{T_1} + \frac{q_1}{T_2} = 0,$$

$$nR \ln \frac{V_2}{V_1} + nR \ln \frac{V_4}{V_3} = 0,$$

$$nR \ln \frac{V_2 V_4}{V_1 V_3} = 0$$

8. Ans (1)

Reaction used in the question



PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (A,B,C,D)

$$\frac{1}{x} + \frac{1}{y} = 1$$

$$\Rightarrow x + y = xy$$

$$\Rightarrow (x-1)(y-1) = 1$$

$$\Rightarrow x = 2, y = 2$$

2. **Ans (A,B)**

$$\begin{aligned}
 (1-x^3)^n &= \sum_{r=0}^n \frac{a_r x^r (1-x)^{3n}}{(1-x)^{2r}} \\
 \Rightarrow \left(\frac{1-x^3}{(1-x)^3} \right)^n &= \sum_{r=0}^n a_r \left(\frac{x}{(1-x)^2} \right)^r \\
 \Rightarrow \left(\frac{1+x+x^2-3x+3x}{(1-x)^2} \right)^n &= \sum_{r=0}^n a_r \left\{ \frac{x}{(1-x)^2} \right\}^r \\
 \Rightarrow \left(1 + \frac{3x}{(1-x)^2} \right)^n &= \sum_{r=0}^n a_r \left\{ \frac{x}{(1-x)^2} \right\}^r \\
 \Rightarrow \sum_{r=0}^n {}^nC_r 3^r \left(\frac{x}{(1-x)^2} \right)^r &= \sum_{r=0}^n a_r \left\{ \frac{x}{(1-x)^2} \right\}^r \\
 \Rightarrow a_r &= {}^nC_r 3^r
 \end{aligned}$$

3. **Ans (A,C,D)**

$S(n, k) \equiv$ No. of ways to partition n distinct items into k parts.

First consider one particular element a_1 forming one group and remaining $n-1$.

elements forming $k-1$ partitions.

Second, consider $n-1$ elements forming k partitions and a , element going in one of ' k ' groups.

$$\text{Hence, } S(n, k) = S(n-1, k-1) + kS(n-1, k)$$

$$\text{No. of onto functions} = k!S(n, k)$$

$$= [k^{n-k} C_1 (k-1)^{n-k} C_2 (k-2)^2 \dots]$$

$$S(5, 3) = \frac{1}{3!} [3^{5-3} C_1 2^{5-3} C_2 1^5] = 25$$

$$\text{No. of onto functions from 6 elements to 3 elements} = 3!S(6, 3)$$

4. **Ans (A,C,D)**

Any point on the given line will be

$$(2\lambda+1, -3\lambda-1, \lambda) \equiv (\alpha, \beta, \gamma)$$

$$\therefore \alpha + \beta + \gamma = 0$$

$$\text{Now } T_{r+1} = {}^9C_r (2x^2)^{9-r} (-x)^{-r}$$

$$= (-1)^r {}^9C_r (2x^2)^{9-r} x^{18-3r}$$

$$18-3r = 0 \Rightarrow r = 6$$

$$\therefore \text{Coefficient of } x^0 = {}^9C_6 2^3$$

$$= \frac{9 \times 8 \times 7}{6} \times 8 = 84$$

5. **Ans (A,B,C,D)**

$$x\vec{a} + y\vec{b} - \vec{c} = \vec{0}$$

$$\Rightarrow \text{If } x + y - 1 = 0$$

then C lies on line segment AB .

If $x > 0$, $y > 0$ and $x + y - 1 = 0$ then C lies segment AB .

If $x < 0$, $y > 0$ and $x + y - 1 = 0$ then C lies on line but not on line segment AB .

If $x > 0$, $y < 0$ and $x + y - 1 = 0$ then C lies on line but not on line segment

If $x > 0$, $y > 0$ and $x + y - 1 < 0$ the C lie inside $\triangle AOB$

6. **Ans (A,C)**

$$A + B^T = B^2$$

$$B + A^T = A^2$$

$$\Rightarrow (A - A^T) - (B - B^T) = B^2 - A^2$$

$$\Rightarrow B^2 - A^2 \text{ is skew symmetric matrix of odd order}$$

$$|B^2 - A^2| = 0$$

$$\text{Now take transpose of equation } B^T = B^2 - A$$

$$\Rightarrow B = (B^T)^2 - A^T$$

$$\Rightarrow (B^T)^2 - B = A^T$$

$$\Rightarrow (B^T)^2 - B = A^2 - B$$

$$\Rightarrow (B^T)^2 - B = A^2 - B$$

$$\Rightarrow (B^T)^2 = A^2$$

$$\Rightarrow |B| = \pm |A|$$

$$\Rightarrow (B^T)^2 - B = (B^2 - B^T)^2 - B$$

$$\Rightarrow B^4 - B^2 B^T - B^T B^2 = 0$$

$$\Rightarrow B^2 (B^2 - B^T) = B^T B^2$$

$$\Rightarrow |B^2| |B^2 - B^T| = |B|^3$$

$$\Rightarrow |B|^2 |A| = |B|^3, \text{ similarly } |A|^2 |B| = |A|^3$$

$$\Rightarrow \text{If } |B| \neq 0 \text{ then } |A| = |B| \text{ and if } |B| = 0 \text{ then } |A| = 0$$

$$\Rightarrow |A| = |B|$$

PART-3 : MATHEMATICS

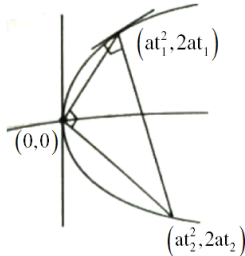
SECTION-I (ii)

7. Ans (A)

(P) For line to become tangent to ellipse

$$a^2 = 2 + 1 \Rightarrow a = \pm\sqrt{3}$$

(Q)



$$\frac{2}{t_1} \cdot \frac{2}{t_2} = -1, t_1 t_2 = -4$$

$$t_2 = -t_1 - \frac{2}{t_1} = -\frac{4}{t_1}$$

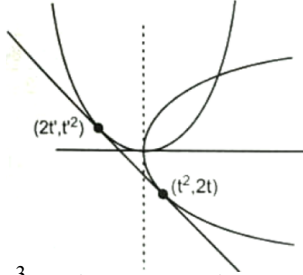
$$m = -t_1 = \pm\sqrt{2}$$

$$(R) ty = x + t^2 \Rightarrow ty - x = t^2 \quad \text{--- (1)}$$

$$t'x = y + t'^2 \Rightarrow t'x - y = t'^2 \quad \text{--- (2)}$$

(1) and (2) are identical

$$\Rightarrow -t = -\frac{1}{t} = \frac{t^2}{t^2} = t^4$$



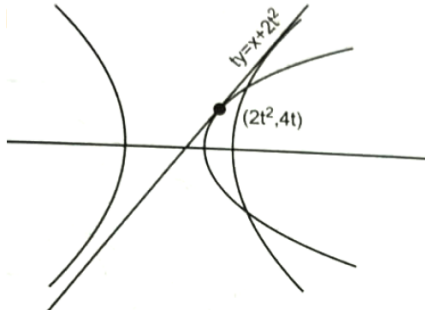
$$t^3 = -1 \Rightarrow t = -1$$

$$-y - x = 1$$

$$x + y + 1 = 0 \Rightarrow k = \sqrt{3}$$

$$(S) ty = x + 2t^2 \quad y = \frac{1}{t}x + 2t$$

For line to become tangent to hyperbola.



$$4t^2 = \frac{1}{t^2} - 3 \Rightarrow 4t^4 + 3t^2 - 1 = 0$$

$$\Rightarrow (4t^2 - 1)(t^2 + 1) = 0$$

$$t = \pm \frac{1}{2} \Rightarrow \pm \frac{1}{2}y = x + \frac{1}{2}$$

$$\Rightarrow 2x + 1 = \pm y \Rightarrow 2x \pm y + 1 = 0$$

8. Ans (B)

(P)

$$P(3 \cos \theta, 2 \sin \theta) \equiv \left(\frac{3(1 - \tan^2 \theta)}{(1 + \tan^2 \frac{\theta}{2})}, \frac{4 \tan \frac{\theta}{2}}{(1 + \tan^2 \frac{\theta}{2})} \right)$$

$\tan \frac{\theta}{2}$ can take infinite rational values

(Q) $P(\pm 3, 0), (0, \pm 2)$

(R) $P(\sqrt{3} \cos \theta, \sin \theta)$

$$\frac{\tan \theta}{\sqrt{3}} = k \in \mathbb{Q}$$

$$\Rightarrow P\left(\frac{\sqrt{3}}{\sqrt{1+3k^2}}, \frac{\sqrt{3}k}{\sqrt{1+3k^2}}\right)$$

$$\text{is } \frac{\sqrt{1+3k^2}}{\sqrt{3}} = \lambda \quad \lambda \in \mathbb{Q}$$

$$\lambda^2 - k^2 = \frac{1}{3}$$

Which can be expressed as product of two rational in infinite ways say for example

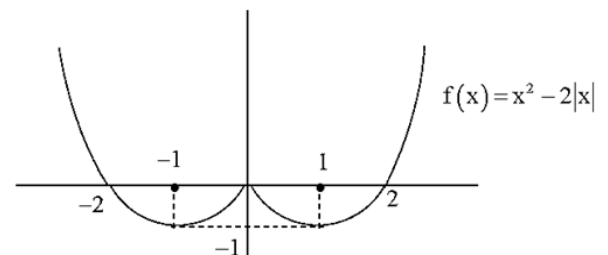
$$\lambda - k = \frac{1}{2};$$

$$\lambda + k = \frac{2}{3}$$

$$\Rightarrow \lambda = \frac{7}{12}, k = \frac{1}{12}$$

(S) $P \equiv (0, \pm 1)$

10. Ans (D)



$$g(x) = \begin{cases} f(x) = x^2 + 2x; & -2 \leq x \leq 1 \\ -1 & -1 < x < 0 \\ 0 & 0 \leq x \leq 2 \\ f(x) = x^2 - 2x; & 2 < x \leq 3 \end{cases}$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (87)

If child buys pencil on the first day then the remaining $n-1$ rupees can be spend in S_{n-1} ways and if he buys pen on the first day then the remaining $n-2$ rupees can be spend in S_{n-2} ways for book on the first day.

$\therefore$ the recurrence relation is

$$S_n = S_{n-1} + S_{n-2} + S_{n-2}$$

$$\therefore S_n = S_{n-1} + 2S_{n-2}$$

$$\text{Clearly } S_1 = 1, S_2 = 3 \{(1,1), 2, 2\}$$

$$\text{Now, } S_3 = S_2 + 2S_1 = 5$$

$$S_4 = 5 + 2 \cdot 3 = 11$$

$$S_5 = 11 + 2 \times 5 = 21$$

$$S_6 = 21 + 22 = 43$$

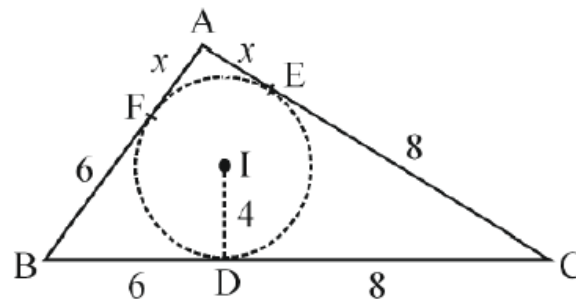
$$S_7 = 43 + 2 \times 21$$

$$= 43 + 42$$

$$= 85$$

$$\therefore \left(\frac{S_n - S_{n-1}}{S_{n-2}} + S_7 \right) = 85 + 2 = 87$$

2. Ans (17.12 or 17.13)



$$s = x + 14$$

$$\Delta = \sqrt{s(s-a)(s-b)(s-c)}$$

$$\Delta = \sqrt{(x+14)(x)(8)(6)} \quad \because \Delta = r \times s$$

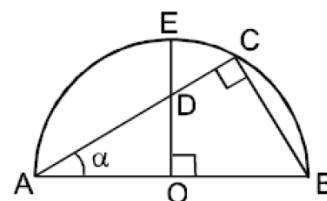
$$\sqrt{8 \times 6x(x+14)} = 4(x+14)$$

$$\Rightarrow x = 7 \Rightarrow \Delta = 84$$

$$\sqrt{\Delta - 3} = 9$$

$$R = \frac{abc}{4\Delta} = \frac{65}{8} = 8.125$$

3. Ans (3)



$$1 + 2 \cos \alpha - \frac{1}{\cos \alpha}$$

$$= \tan \alpha + 2 \sin \alpha$$

$$\Rightarrow \cos \alpha + \cos 2\alpha = \sin \alpha + \sin 2\alpha \left(\cos \frac{\alpha}{2} \neq 0 \right)$$

$$\Rightarrow \tan \frac{3\alpha}{2} = 1 \Rightarrow \alpha = \frac{\pi}{6}$$

4. Ans (2.23 or 2.24)

Since $z_1 + z_2 + z_3 = 1$

$$\Rightarrow z_1 + z_2 + z_3 = 1$$

$$\Rightarrow \frac{1}{z_1} + \frac{1}{z_2} + \frac{1}{z_3} = 1$$

$$\Rightarrow z_2 z_3 + z_3 z_1 + z_1 z_2 = 1$$

Now, consider a cubic equation having roots

z_1, z_2, z_3

$$(z - z_1)(z - z_2)(z - z_3) = z^3 - (z_1 + z_2 + z_3)z^2 +$$

$$(z_1 z_2 + z_2 z_3 + z_3 z_1)z - z_1 z_2 z_3 = 0$$

$$\Rightarrow z^3 - z^2 + z - 1 = 0$$

$$\Rightarrow z^2(z - 1) + 1(z - 1) = 0$$

$$\Rightarrow z = 1, z = \pm i$$

$\therefore$ Roots $z_1 = -i, z_2 = 1, z_3 = i$

$$|z_1 + z_2^2 + z_3^3| = \sqrt{5}$$

5. Ans (2.14)

$$\sum_{n=0}^{\infty} \tan^{-1} \left(\frac{\tan^{-1} \left(\frac{(n+2)-(n+1)}{1+(n+1)(n+2)} \right)}{1 + \tan^{-1} \left(\frac{1}{n+1} \right) \tan^{-1} \left(\frac{1}{n+2} \right)} \right)$$

$$\sum_{n=0}^{\infty} \tan^{-1} \left(\frac{\tan^{-1} \left(\frac{1}{n+1} \right) - \tan^{-1} \left(\frac{1}{n+2} \right)}{1 + \tan^{-1} \left(\frac{1}{n+1} \right) \cdot \tan^{-1} \left(\frac{1}{n+2} \right)} \right)$$

$$\sum_{n=0}^{\infty} \left(\tan^{-1} \left(\tan^{-1} \left(\frac{1}{n+1} \right) \right) \right) - \tan^{-1} \left(\tan^{-1} \left(\frac{1}{n+2} \right) \right)$$

$$\tan^{-1}(\tan^{-1}(1)) = \tan^{-1} \left(\frac{\pi}{4} \right)$$

$$\therefore p = \pi$$

$$\cos^{-1}(\cos(\pi - 1)) = \pi - 1 = 2.14$$

6. Ans (7)

$$\text{Given } f(x+5) \geq f(x) + 5 \quad \text{---(1)}$$

$$f(x+1) \leq f(x) + 1 \quad \text{---(2)}$$

From (1) and (2) we can write

$$f(x) + 5 \leq f(x+5) \leq f(x+4) + 1$$

$$\leq f(x+3) + 2$$

$$\leq f(x+2) + 3$$

$$\leq f(x+1) + 4$$

$$\leq f(x) + 5$$

$\therefore$ Equality exist every where

$$\therefore f(x+1) = f(x) + 1 \quad \therefore f(1) = 1$$

$$\Rightarrow f(2) = 2, f(3) = 3 \dots\dots\dots f(2023) = 2023.$$

$$\text{Now } g(x) = f(x) + 7 - x \Rightarrow g(2023) = 7$$

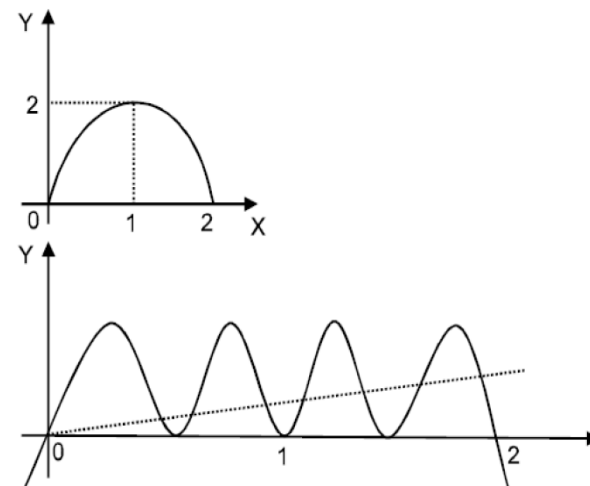
7. Ans (16)

$$f(x) = 2x(2 - x)$$

$$y = f(f(f(x))) ; 0 \leq x \leq 2$$

$$y = 32x(2 - x)(x - 1)^2(2x^2 - 4x + 1)^2$$

$$; 0 \leq x \leq 2$$



8. Ans (4)

Equation of tangent at (x_i, y_i) on the curve $y = f(x)$

$$\text{is: } Y - y_i = m(X - x_i)$$

$$\text{Put } Y = 0, X = x_{i+1},$$

$$\text{then } 0 - y_i = m(x_{i+1} - x_i)$$

$$\Rightarrow x_{i+1} - x_i = -\frac{y_i}{m}$$

$$\therefore \frac{x_{i+1}}{x_i} - 1 = -\frac{y_i}{mx_i}$$

If $x_1, x_2, x_3, \dots, x_n$ form G.P.

then $r - 1 = -\frac{y_i}{mx_i}$ (Where r is common ratio of G.P.)

$$\therefore m = \frac{dy}{dx} = -\frac{y}{(r-1)x}$$

$$\Rightarrow \frac{dy}{y} + \frac{dx}{(r-1)x} = 0$$

$$\Rightarrow \ell n y + \left(\frac{1}{r-1} \right) \ell n x = k$$

$$\Rightarrow y(x)^{\frac{1}{r-1}} = c$$

$$\Rightarrow r = 2, xy = c, e = \sqrt{2}$$